

A Range-Complemented Counterexample to a Compatibility Question

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Source Question

Castillo, Cuellar, Ferenczi, and Moreno ask in Problem 3 of *Complex structures on twisted Hilbert spaces* [1]:

Assume $\Omega : \ell_2 \rightarrow \ell_2$ is a singular quasi-linear map, and that T, U are bounded linear operators on ℓ_2 such that (T, U) is compatible with Ω . Does it follow that $T - U$ is compact?

Here compatibility means $T\Omega \equiv \Omega U$, i.e. the difference is the sum of a bounded homogeneous map and a linear map. We use the standard equivalent formulation that a quasi-linear map is singular when it is nontrivial on every infinite-dimensional subspace of its domain.

Idea of the Proof

The compactness conclusion is false for a very simple reason: the statement does not require the range of Ω to be essential in the target copy of ℓ_2 . Put a singular quasi-linear map into a complemented half of ℓ_2 . The projection onto the other half annihilates Ω , hence is compatible with the zero operator on the domain, but that projection is not compact.

Counterexample

Theorem 1. *Problem 3 has a negative answer as stated. There exist a singular quasi-linear map $\Omega : \ell_2 \rightarrow \ell_2$ and bounded operators $T, U : \ell_2 \rightarrow \ell_2$ such that (T, U) is compatible with Ω , while $T - U$ is not compact.*

Proof. Let $\mathcal{K} : \ell_2 \rightarrow \ell_2$ be the Kalton–Peck quasi-linear map

$$\mathcal{K}(x) = x \log \frac{|x|}{\|x\|_2},$$

with the usual coordinatewise interpretation and $\mathcal{K}(0) = 0$. This is a singular quasi-linear map, as recalled in the source paper before Corollary 4.6.

Decompose ℓ_2 as the direct sum of its even and odd coordinate subspaces,

$$\ell_2 = H_{\text{even}} \oplus H_{\text{odd}}.$$

Let $J : \ell_2 \rightarrow H_{\text{even}}$ be the isometric embedding $Je_n = e_{2n}$, and let $R : \ell_2 \rightarrow \ell_2$ be the bounded left inverse defined by $Re_{2n} = e_n$ and $Re_{2n-1} = 0$. Define

$$\Omega = J\mathcal{K} : \ell_2 \rightarrow \ell_2.$$

Since J is bounded linear and $\mathcal{K}$ is quasi-linear, Ω is quasi-linear.

We first check that Ω is singular. Suppose that Ω were trivial on an infinite-dimensional subspace $E \subset \ell_2$. Then for some linear map $L : E \rightarrow \ell_2$ and bounded homogeneous map $B : E \rightarrow \ell_2$ we would have $\Omega|_E = L + B$. Applying R gives

$$\mathcal{K}|_E = R\Omega|_E = RL + RB,$$

where RL is linear and RB is bounded homogeneous. Thus $\mathcal{K}$ would be trivial on E , contradicting the singularity of the Kalton–Peck map. Therefore Ω is singular.

Let $T = P_{\text{odd}}$ be the coordinate projection onto H_{odd} , and let $U = 0$. Then

$$T\Omega = P_{\text{odd}}J\mathcal{K} = 0 \quad \text{and} \quad \Omega U = \Omega 0 = 0.$$

Hence (T, U) is compatible with Ω , even with equality rather than just equivalence.

Finally, $T - U = P_{\text{odd}}$ is not compact: it fixes the orthonormal sequence $(e_{2n-1})_{n \geq 1}$, which has no norm-convergent subsequence. This completes the counterexample. $\square$

Verification Notes

The proof uses only two standard facts: the Kalton–Peck map is singular, and singularity is preserved by postcomposition with an isomorphic embedding whose range has a bounded left inverse. The construction does not rely on any unproved property of Z_2 .

Scope and Limitations

This answers only the arbitrary singular quasi-linear-map Problem 3. The construction is deliberately not a centralizer counterexample, and it does not address the subsequent source questions about exact sequences representing Z_2 , singular centralizers on ℓ_2 , compatible complex structures on hyperplanes of derived spaces, or uniqueness of complex structure on Z_2 .

Novelty Check

The run indexes were searched for the arXiv id, title, Kalton–Peck, singular quasi-linear maps, compatible pairs, and compactness of $T - U$. No duplicate packet or attempt for this source problem was found. Bounded web searches for exact phrases from the problem and for **Kalton–Peck compatible T–U compact** did not surface a later answer or the range-complemented counterexample recorded here.

References

- [1] J. M. F. Castillo, W. Cuellar, V. Ferenczi, and Y. Moreno, *Complex structures on twisted Hilbert spaces*, arXiv:1511.05867.
- [2] N. J. Kalton and N. T. Peck, *Twisted sums of sequence spaces and the three space problem*, Transactions of the American Mathematical Society 255 (1979), 1–30.